

ROSHAN TANEJA

Machine Learning + Geospatial Data Scientist

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COMMUNITY AND RESEARCH AWARDS

- o AICC eurlPS Poster: Copenhagen 2025
- o NeurIPS Winner: ML for Social Impact 2024
- o ML4EO Presenter 2024
- o Intl. Humanitarian Services Award 2024
- o Nominee, Intl. Children's Peace Prize 2024
- o United Nations (NGFP) SDG #6 Award 2023

ACADEMIC EXCELLENCE AWARDS

- o Computer Science Department Honor 2025
- o AP Scholar with Distinction 2024
- o Featured: Harvard Spatial Data Lab 2024
- o Rotary Computer Science Award 2023
- o Robotics Regionals 2023, 2024, 2025

LEADERSHIP

- o Youth President, Maji Wells 2021-Present
- o President/Spirit Chair, SHP 2023-2025
- o Lead, Robotics, SHP 2022-2025

OPEN SOURCE PROJECTS

- o SneakerArbitrage 2024-2025
- o PestoFTC 2025
- o Cost Of Colonialism Data Sheets 2024-2025
- o rpi_f1_terminal_hud 2024-2025
- o Dawg-App-Platform 2024

SOFTWARE SKILLS

- o Python o C++ o Appscript o OCR o Docker
- o Embedded Search o YOLOvX/CNNs/RetinaNet
- o TransferLearning o Object Detection o Linux
- o Full Stack Development o Web Apps

HARDWARE SKILLS

- o CAM/CAD (F360, SW) o Hardware Design
- o PCB Design o IC Design o ESP32 Software
- o Systems Engineering

GEOSPATIAL SKILLS

- o ArcGIS Pro o Google Earth Engine o PostGIS
- o AlphaEarth o Geopandas o Labeling Pipelines
- o MCDA o SOAR o mAP/F1

LINKS

- 🌐 roshan.codes
- 🐙 github.com/roshantaneja
- 🌐 linkedin.com/in/roshan-taneja
- 🔗 roshantaneja.stck.me

EDUCATION

EECS @ UC Berkeley 2025-2029

- o Structure & Interpretation of Computer Programs
- o Multivariable Calculus
- o Introduction to Amateur Rocketry
- o Data Structures & Algorithms
- o Linear Algebra
- o Discrete Mathematics & Probability Theory
- o Foundations of Signals, Dynamical Systems, & Information Processing
- o Introduction to Circuits & Devices
- o Space Technologies and Rocketry Club

HIGH SCHOOL DIPLOMA @ Sacred Heart Preparatory, Atherton 2021-2025

- o GPA: 3.98-ACT: 36
- o AP: Calc BC, Phys 1, CompSci A, Macro, Micro, Phys C, Eng Lang, Eng Lit, Spanish
- o Post-AP: Math, CompSci, Machine Learning, Senior Honors Independent Study

SPATIAL RESEARCH CERTIFICATE @ Harvard GIS Institute Spring 2025

INTERMEDIATE DATA SCIENCE @ UCLA Summer 2024

REMOTE SENSING @ MIT Beaver Works Spring 2024

PROJECTS & WORK EXPERIENCE

AVIONICS MEMBER @ Cal Space Technologies and Rocketry 2025-Present

- o Producing avionics systems for high-power rocketry (30,000 ft), custom PCBs using Altium Designer, Soldering and testing high and low voltage boards, volume and mass constraint boards with microcontrollers, ESP32 Code

DATA SCIENCE RESEARCHER @ Cost of Colonialism (COCO) 2024-2025

- o Created OCR+NLP semantic search over 1600s-1800s shipping manifests; embedding-based retrieval + query UI for historians. Automated entity extraction across noisy scans; improved searchability, reducing manual review time. Identified key impacts of Dutch East India Company and British Colonial rule on Indian economy and growth.

RESEARCH INTERN @ Harvard University 2024-2025

- o Working with Prof. Wang in the Center for Geographic Analysis (CGA) to expand use cases for placement of water harvesting units using satellite imagery & machine learning for Maasai in Tanzania. (N=10k+ tiles)

YOUTH PRESIDENT, WATER PROGRAMS @ Maji Wells 2021-2025

- o Developed a machine learning-based satellite image classification model to identify living locations of Maasai to plan the placement of water harvesting units and reservoirs. **Fundraised \$300k** to deploy 500 rainwater harvesting solutions serving **30,000 Maasai**.

ENGINEERING INTERN @ Fluxon Apps 2023

- o Built a Typescript tool to automate previously manual billing for employees. Saved 2000 annual man-hours; first intern for the company.

MECHANIC @ Calabazas Cyclery 2022-2023

- o Fixed bikes, created website and deployed new inventory management system, online catalog.

PUBLICATIONS, PATENTS & ACADEMIC CONFERENCES

PRESENTER: NEURIPS 2024 WINNER: "MACHINE LEARNING 4 SOCIAL IMPACT" 2024

- o 1 of 4 Winners from 330 entries, 14,000 attendees.

PROVISIONAL PATENT APPROVED, PUBLICATION: OBJECT DETECTION IN SATELLITE IMAGERY FOR OPTIMIZING THE PLACEMENT OF RAINWATER UNITS 2024

- o US Patent 63/703,232. Published in NHSJS.

PRESENTER: MACHINE LEARNING 4 EARTH OBSERVATION CONFERENCE @ University of Exeter 2024

PUBLICATION: IMPACT OF RAINWATER HARVESTING ON 4500+ MAASAI 2024